

FUTURE_CS_02 | Cyber Security Internship

PHISHING EMAIL DETECTION & AWARENESS REPORT

Task 2 — Email Threat Analysis

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DATE

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TOOL USED

MXToolbox Analyzer

TRACK

Future Interns — Cyber Security

PHISHING: 3

SUSPICIOUS: 2

SAFE: 1

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Executive Summary

A phishing email detection analysis was conducted on **6 email samples** using **MXToolbox Email Header Analyzer**. Each email was examined for SPF, DKIM, and DMARC authentication results and classified as SAFE, SUSPICIOUS, or PHISHING.



Verdict	Count	Percentage	Risk Level
PHISHING	3	50%	Critical — Immediate threat
SUSPICIOUS	2	33%	Medium — Requires verification
SAFE	1	17%	Low — Legitimate email
TOTAL	6	100%	—

Key Finding

5 out of 6 emails (83%) showed some form of security concern. The most common indicators were **SPF failure, missing DKIM signatures, and fake sender domains**. Even a legitimate email (Future Interns offer letter) showed poor email security configuration.

What Is Phishing?

Phishing is a cyber attack where criminals send fake emails pretending to be trusted organizations such as banks, tech companies, or government agencies to steal login credentials, money, or personal information from victims.

Common Phishing Techniques

#	Technique	How It Works
1	Sender Spoofing	Display name looks real but actual email domain is completely different
2	Urgency Tactics	"Your account will be blocked!" — forces victims to act without thinking
3	Fake Domains	amazon-orders-alert.xyz used instead of the real amazon.com
4	Reply-To Hijack	Replies secretly redirected to attacker-controlled Gmail accounts
5	Bulk Email Tools	Special software sends thousands of phishing emails simultaneously
6	Brand Impersonation	Official logos and branding copied from PayPal, Amazon, banks
7	Suspicious TLDs	.tk, .ml, .xyz domains are cheap and heavily used by phishers

Email Authentication — What It Means

Check	What It Verifies	PASS	FAIL
SPF	Is the sending server authorized?	Legitimate server	Unauthorized server
DKIM	Was the email tampered with?	Genuine signature	No/invalid signature
DMARC	Do both SPF & DKIM pass?	Email is legitimate	Suspicious / phishing
Alignment	Does display name match sender?	Names match	Name is spoofed

Email Analysis Results

EMAIL 1

SUSPICIOUS

Internship Selection Confirmation | Cyber Security

From: contact@futureinterns.com

SPF

PASS (not aligned)

DKIM

FAIL — No signature

DMARC

FAIL — Policy: none

SPF passes via Google servers but DKIM is completely missing and DMARC policy is not enforced. Shows poor email security configuration even from a legitimate organization.

EMAIL 2

PHISHING

URGENT: Your PayPal account suspended

From: security@paypal-accounts-alert.tk | Reply-To: collect@gmail.com

SPF

FAIL

DKIM

FAIL

DMARC

FAIL

All checks failed. Fake .tk domain used to impersonate PayPal. Reply-To points to attacker Gmail. Classic brand impersonation with urgency language.

EMAIL 3

SAFE

Your pull request was merged

From: noreply@github.com

SPF

PASS

DKIM

PASS

DMARC

PASS

All three authentication checks passed. Sent from official GitHub servers. No suspicious indicators detected. This is what a legitimate email looks like.

Email Analysis Results (Continued)

EMAIL 4 **PHISHING** Your Amazon order cancelled - Action Required

From: support@amazon-orders-alert.xyz | Reply-To: refund@gmail.com

SPF
FAIL

DKIM
FAIL

DMARC
FAIL

All checks failed. amazon-orders-alert.xyz is NOT amazon.com. Reply-To redirects to attacker Gmail. Uses "Action Required" urgency to trick victims.

EMAIL 5 **PHISHING** URGENT: Your bank account is blocked

From: alert@hdfcbank-secure-login.ml | X-Mailer: Bulk Email Sender v2.1

SPF
FAIL

DKIM
FAIL

DMARC
FAIL

All checks failed. Fake .ml domain impersonating HDFC Bank. Uses Bulk Email Sender v2.1 — a mass phishing tool. Extreme urgency language used to panic victims.

EMAIL 6 **SUSPICIOUS** You have been selected for a job opportunity!

From: jobs@career-opportunity-global.xyz | Reply-To: hr.recruit2026@gmail.com

SPF
SOFTFAIL

DKIM
NONE

DMARC
NONE

SPF softfail = partially unauthorized sender. No DKIM or DMARC configured. Suspicious .xyz domain. Gmail used for HR replies — not professional for a real company.

Prevention & Awareness Guidelines

Follow these 10 best practices to protect yourself and your organization from phishing attacks:

- 1 Always check the sender's **actual email domain** — not just the display name. PayPal emails come from @paypal.com, not @paypal-alert.tk
- 2 **Hover over links** before clicking to reveal the true destination URL in the bottom of your browser
- 3 Legitimate companies **NEVER ask for passwords**, credit card numbers, or OTPs via email
- 4 If an email creates **urgency or fear** — pause and verify directly with the company before acting
- 5 Use **MXToolbox.com/EmailHeaders.aspx** to check SPF, DKIM, and DMARC on suspicious emails
- 6 **Report suspicious emails** to your IT/security team immediately — do not forward them to others
- 7 Enable **Multi-Factor Authentication (MFA)** on all accounts to limit phishing damage even if credentials are stolen
- 8 Use a **password manager** — it won't autofill credentials on fake phishing sites
- 9 **Never open attachments** from unknown senders, especially .exe, .zip, .docm, or .xlsm files
- 10 When in doubt — **call the company directly** using the number from their official website, not from the email

Quick Red Flag Reference

Red Flag	What It Means
Sender domain mismatch	Display name differs from actual sending domain — spoofing attempt
SPF / DKIM / DMARC FAIL	Email is unauthorized or has been tampered with
Reply-To is Gmail	Attacker wants your reply to go to their personal inbox
.tk / .ml / .xyz domain	Cheap domains heavily used for phishing campaigns
URGENT in subject line	Psychological pressure to bypass rational thinking

Bulk Email Sender
header

Mass phishing campaign — not a personal email

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Conclusion

Out of **6 emails analyzed**, **3 were confirmed PHISHING**, **2 were SUSPICIOUS**, and **1 was SAFE**. This clearly demonstrates that email-based threats are widespread and that authentication protocols are reliable detection tools.

A key finding is that even the legitimate **Future Interns offer letter** showed email security misconfigurations — missing DKIM and DMARC policy set to "none". This highlights that proper email security setup is critical for all organizations.

Users should regularly verify suspicious emails, enable MFA, and stay educated about phishing indicators to significantly reduce risk.

Metric	Value
Total Emails Analyzed	6
Phishing Confirmed	3 (50%)
Suspicious	2 (33%)
Safe	1 (17%)
Tool Used	MXToolbox Email Header Analyzer
Key Phishing Indicator	SPF + DKIM + DMARC all failing = confirmed phishing
Best Prevention	MFA + Email header verification + Domain checking

Disclaimer: This report was prepared as part of the Future Interns Cyber Security Internship.

All phishing email samples are used for educational purposes only. No actual attacks were performed.

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